

# Vedant Singh

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🔗 Leetcode

## PROFESSIONAL SUMMARY

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**Microsoft Certified Azure AI Associate** and BTech student at VIT Chennai (**CGPA 9.14**) specializing in AI-integrated full-stack development. Proven technical problem-solver with a 2nd rank in the NetSim Hackathon, dedicated to bridging the gap between advanced Generative AI models and scalable, user-centric web applications.

## EDUCATION

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**Btech in Electronics and Computer Engineering**  
VIT Chennai  
CGPA 9.14

August 2023 – Present

## SKILLS

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### Programming Languages

Python, C/C++, Java, JavaScript, OracleSQL/MySQL

### Data Science

Pandas & NumPy, Scikit-learn, Seaborn, Statistical Analysis(R), LIME & SHAP

### AI & Machine Learning

Machine Learning, Deep Learning, NLP, NLTK

### Web, Cloud & Tools

HTML/CSS, React & NodeJS, Streamlit, Microsoft Azure, AWS, Git & Figma, REST APIs

## PROJECTS

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### Real-Time Sentiment Analysis Web App 📄

Developed a responsive web application using **Streamlit** to provide real-time sentiment predictions, successfully deploying it to the cloud for public access. Engineered a robust NLP preprocessing pipeline using **NLTK** and **TF-IDF** vectorization, integrating custom rule-based enhancements for idiom detection and negation patterns. Trained and optimized a **Linear SVM** classifier on nearly 150,000 multi-domain reviews, achieving an 82.17% accuracy rate to deliver highly contextual and accurate opinion analysis.

### Cronos AI | Personalized AI Chatbot

Developed a hybrid voice assistant in **Python** combining low-latency offline hotword detection (**Porcupine**) with cloud-based LLM intelligence (**Gemini API**). Integrated **Google Speech Recognition** to parse voice commands, enabling system-level automation (opening apps/sites) and complex natural language Q&A.

### Diabetes Predictor using Machine Learning

Engineered a machine learning pipeline handling class imbalance with **SMOTE** and missing data via median imputation. Outperformed Logistic Regression (71%) with a **Random Forest** model (77%), validating non-linear modeling for health data. Conducted a comparative XAI study proving **LIME's** mathematical instability; implemented **SHAP** to derive robust, game-theoretic explanations for top drivers (Glucose, BMI).

### Rockfall Prediction System

Engineered an end-to-end full-stack application to classify geological stability zones (Safe/Moderate/Unsafe) using a **Random Forest** classifier. Generated a comprehensive **synthetic dataset** simulating critical environmental factors (soil moisture, slope angle, precipitation) to overcome data scarcity and train the model. Developed a **Flask** backend to process environmental inputs and serve predictions to a responsive frontend dashboard. Implemented real-time data visualization charts to display risk trends, providing actionable safety insights for vulnerable areas.

## CERTIFICATIONS & ACHIEVEMENTS

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### Microsoft Azure AI Associate 📄

Certified by Microsoft, validating expertise in designing and deploying AI solutions on Azure, including integrating AI services for diverse applications

### NetSim Hackathon Rank 2 📄

Designed and simulated networks using NetSim, optimizing network performance parameters and secured 2nd place.